

**SIMATS ENGINEERING
ASSESSMENT TOOL 1****COURSE NAME: CSA07 COURSE CODE: Computer Networks**

COURSE OUTCOME COVERED**CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25% A	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding major misconceptions
Explanation	25% 35	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	20% 5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written difficult to understand

ASSESSMENT TOOL - 1

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Q1) Given ,

File size = 150 MB

Bandwidth = 50 Mbps

Segment size = 1500 bytes

Overhead = 40 bytes per frame

CALCULATIONS :

① Number of segments : Since the segment size is given in bytes , first convert the file size into bytes

1 MB = 10,00,000 Bytes

150 MB = 15,00,000 000 Bytes

Number of segments :
$$\frac{\text{File size}}{\text{Segment size}}$$

$$= \frac{15,00,00,000}{1500} \longrightarrow 1,00,000$$

∴ The transport layer divides the 150 MB file into 1,00,000 segments each of 1500 bytes

Each segment is encapsulated into one frame by Data link

$$\text{Number of frames} = \text{Number of segments} = 1,00,000$$

② Each frame carries 40 bytes of header and trailer

$$\text{Total overhead} = 40 \times 100,000 \rightarrow 40,00,000 \text{ Bytes} \rightarrow 4 \text{ MB}$$

$\therefore$ The additional header and trailer added to every frame result in a total overhead of 4 MB

③ Transmission time = Total Data $\div$ Bandwidth

$$\begin{aligned} \text{Total Data} &= 150 \text{ MB} \times 8 \quad [\because 1 \text{ byte} = 8 \text{ bits}] \\ &= 1200 \text{ Megabits} \end{aligned}$$

$$\text{Transmission time} = 1200 \div 50 \rightarrow 24 \text{ seconds}$$

THEORY :

- The Transport layer divides the file into segments, assigns sequence numbers and ensures reliable end-to-end communication through retransmission and flow control

- The Data link layer converts segments into frames by adding a 40-byte header and trailer for addressing & error detection

- Together, both layers provide accurate, reliable and efficient data transmission

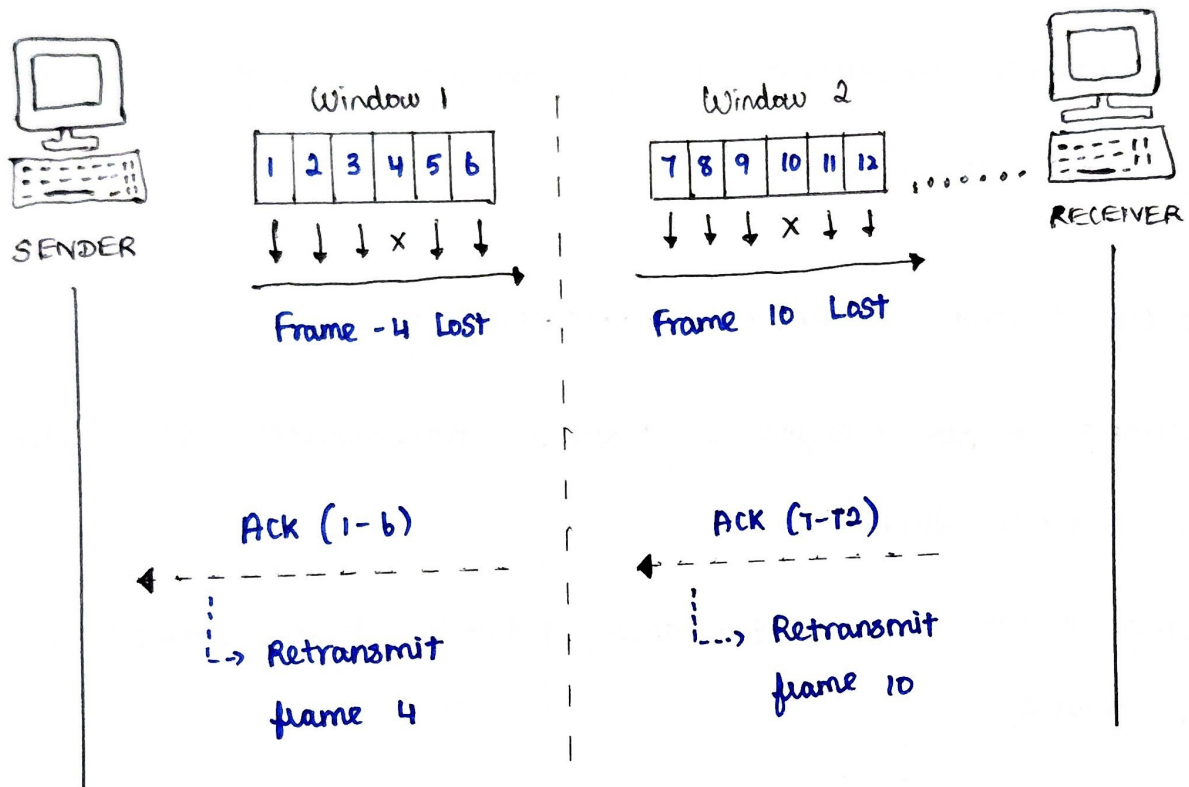
Given,

Window size = 6 Frames

Total frames = 48

Data per frame = 1024 Bytes

• One frame is lost in every window



① Number of windows = Total frames $\div$ Window size

= $48 \div 6 \rightarrow 8$

$\therefore$ The sender sends 48 frames in groups of 6, so 8 transmission windows are required

② Number of Retransmissions = Total windows = 8

Lost frames = 1×8

$\therefore$ Since one frame is lost in each of 8 windows, the sender retransmits 8 frames

THEORY :

- The sliding window Protocol allows the sender to transmit multiple frames before waiting for an acknowledgment (Ack)
- Flow control prevents sender from transmitting data faster than the receiver can process, avoiding data loss and congestion
- It improves communication by ensuring efficient, reliable and orderly data transmission

HOW FLOW CONTROL IMPROVES EFFICIENCY :

- ① Preventing Buffer overflow : Matches transmission speed directly to receiver memory limits
- ② Reducing Packet Drops : Decreases network drops caused by exhausted receiver queues
- ③ Minimizes Retransmissions : Saves network bandwidth by avoiding data-recovery cycles
- ④ Maximizing Link Utilization : Allows continuous frame transmission before receiving an ACK

Q3) Given ,

Original IP packet = 12,000 Bytes

MTU = 1500 Bytes

IP header = 20 Bytes

① Calculate payload in each fragment

$$\text{Payload} = \text{MTU} - \text{Header}$$

$$= 1500 - 20$$

$$\text{Payload} = 1480 \text{ Bytes}$$

② Calculate number of fragments

$$\text{no. of fragments} = \text{total packet size} \div \text{payload per fragment}$$

$$= 12000 \div 1480$$

$$= 8.1 \text{ (roundup)}$$

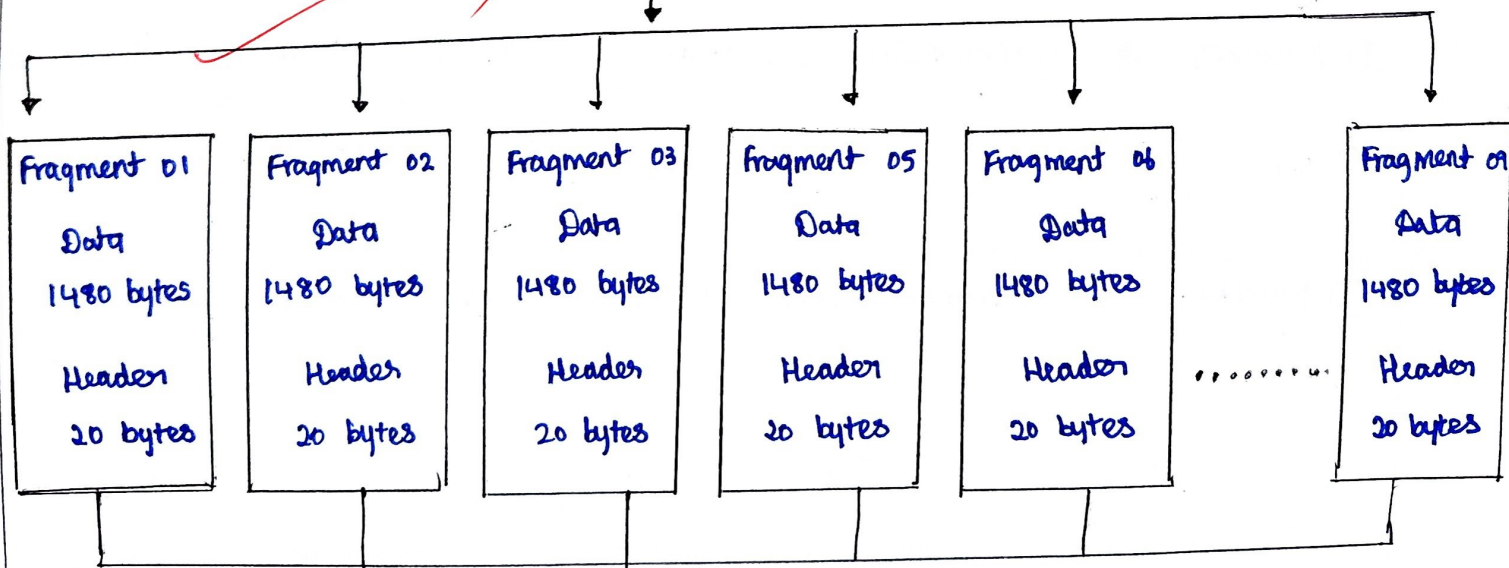
$$\text{No. of fragments} = 9$$

$$\text{③ Calculate Header Overhead} = 9 \times 20 \rightarrow 180 \text{ bytes}$$

original IP packet
12,000 bytes



Router
(MTU = 1500 bytes)



Destination $\rightarrow$ Reassembles all fragments to recover original packet

How does the network layer ensure successful delivery of fragmented packets?

- It divides a large IP packet into smaller fragments based on the MTU of the network
- Each fragment is assigned its own IP header containing the Identification, Fragment offset, and Flags fields
- At the destination, the network layers uses these header fields to identify, arrange, and reassemble all the fragments in the correct order, restoring the original packet

Q4) Given,

• Sliding window size = 6 Frames

Total frames transmitted = 48 Frames

Data in each frame = 1024 bytes

One frame in every window is lost and retransmitted

① Number of transmission windows : $\text{Total frames} \div \text{Window size}$
 $\Rightarrow 48 \div 6 = 8 \text{ windows}$

② Number of Retransmissions : $1 \times 8 \rightarrow 8 \text{ retransmission}$



Sender



Data Flow



Receiver

Window 01

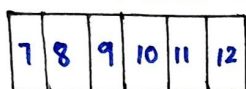


↓ one frame lost
(Retransmitted)

← ACK

← Retransmission

Window 02



↓ one frame lost
(Retransmitted)

← ACK

← Retransmission

⋮

Window 08



↓ One frame lost
(Retransmitted)

← ACK

← Retransmission

How Flow Control improves efficiency :

- Flow control ensures that the sender doesn't send data faster than the receiver can process
- It prevents buffer overflow at the receiver and reduce congestion
- By using a sliding window, the sender transmits only a limited no. of frames and waits for ACK
- Avoids unnecessary retransmissions, saves bandwidth and improves overall communication efficiency

Q5) Given,

Total file size = 600 MB

Checkpoint inserted after every 60 MB

Failure occurs after 390 MB

① No. of checkpoints created before failure

$$= \frac{\text{Data transferred before failure}}{\text{Checkpoint Interval}}$$

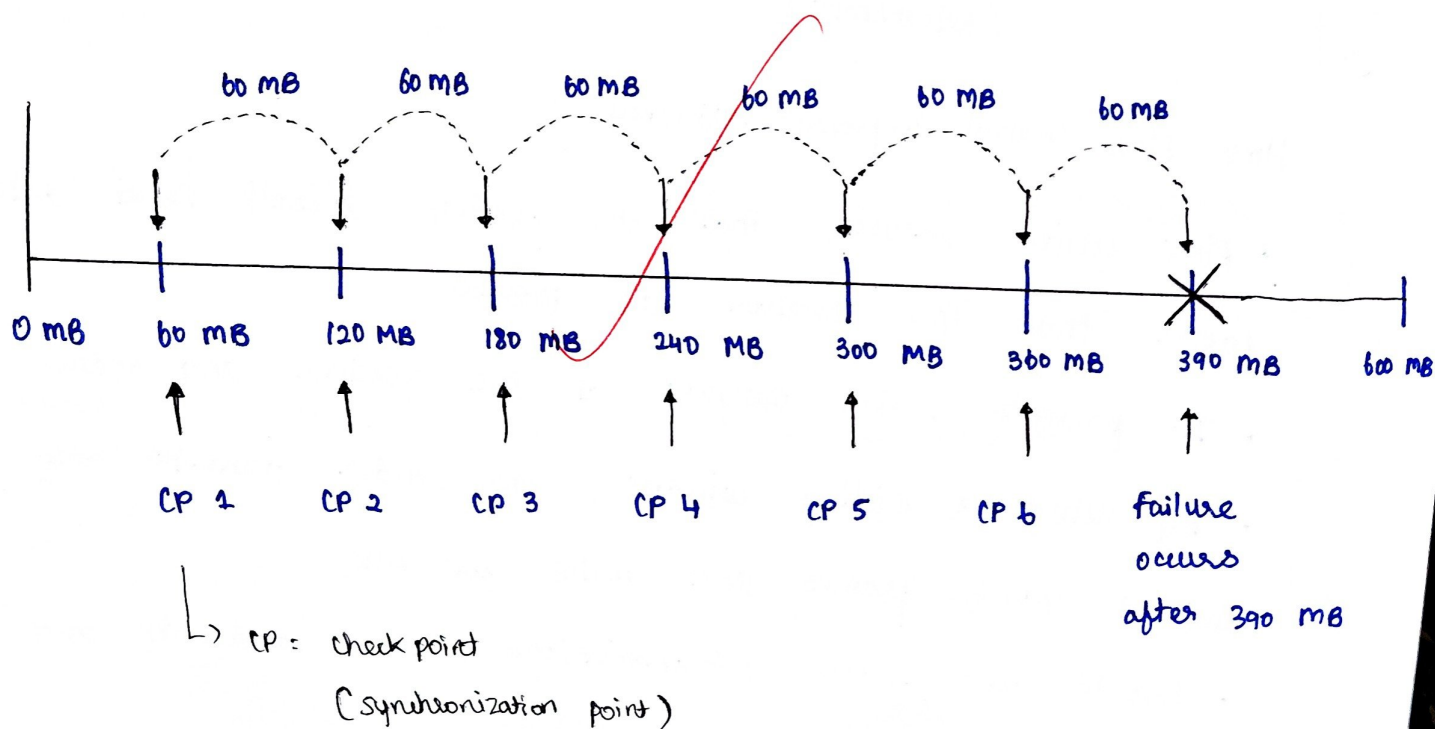
$$= \frac{390}{60} \rightarrow 6 \text{ checkpoints}$$

② Amount of data to be retransmitted after recovery

Last successful checkpoint = 360 MB

Failure occurs after = 390 MB

Data that must be retransmitted = $390 - 360 \rightarrow 30 \text{ MB}$



Importance of Synchronization in Session Layer :

- Provides checkpoints during data transmission for recovery
- Resumes communication from the last checkpoint after a failure instead of restarting the entire transfer
- Saves bandwidth, time and network resources by avoiding retransmission of successfully received data
- Manages communication sessions and controls data exchange between sender and receiver
- Synchronizes multimedia streams (such as audio and video) to ensure smooth playback

Q6) Given,

Original file size = 900 MB

Compression by presentation layer = 40%.

Encryption adds 5% overhead on compressed size

① Compressed file size :

Compression = 40% of 900 MB $\rightarrow$ 360 MB

Compressed size = $900 - 360 \rightarrow 540$ MB

② Encrypted file size :

Encryption overhead = 5% of compressed size

= $0.05 \times 540 \rightarrow 27$ MB

Encrypted size = $540 + 27 \rightarrow 567$ MB

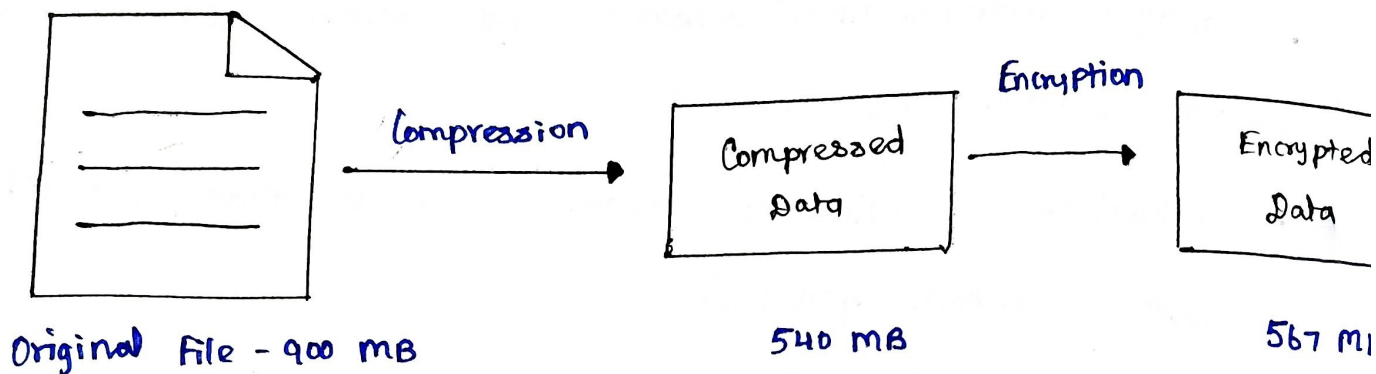
③ Total reduction = $900 - 567 \rightarrow 333 \text{ MB}$

Percentage reduction = $\frac{333}{900} \times 100 \rightarrow 37\%$

$\therefore$ Compressed file size = 540 MB

Encrypted file size = 567 MB

Total % reduction compared to original file = 37%



- IMPROVES TRANSMISSION EFFICIENCY :
 - Compression reduces the size of data before transmission
 - Smaller data size means less bandwidth range
 - It reduces transmission time, network congestion and storage space
- IMPROVES SECURITY :
 - Encryption converts the compressed data into an unreadable format
 - It protects data from unauthorized access, eavesdropping and tampering
 - Ensures confidentiality of data during transmission

Given,

Total data to be transferred = 3 GB

Effective throughput of network = 120 Mbps

Each request size at application layer = 3 MB

(1 Byte = 8 bits, 1 GB = 1024 MB)

① Total no. of requests generated

Total data = 3 GB $\rightarrow 3 \times 1024 \text{ MB} \rightarrow 3072 \text{ MB}$

Each request = 3 MB

No. of requests = $\frac{3072}{3} \rightarrow 1024 \text{ requests}$

② Total transmission time

Total data in bits = 3 GB $\times 8 \text{ bits / Byte}$

= $3 \times 1024 \times 1024 \times 1024 \times 8 \text{ bits}$

= 25,76,98,03,776 bits

Throughput = 120 Mbps $\rightarrow 120 \times 10^6 \text{ bits/sec}$

Time = $\frac{25,76,98,03,776}{120 \times 10^6}$

Time = 214.75 seconds (approx)

$\therefore$ Total no. of requests generated = 1024

Total transmission time = 214.75

HOW APPLICATION LAYER PROVIDES RELIABLE SERVICES :

- ① File Transfer Protocol (FTP) : Uses commands and responds to ensure correct file delivery
- ② Segmentation into Requests : Application layer divides large data into manageable parts (3 MB each)
- ③ Acknowledgment & Response : The receiver sends acknowledgements for each request, and the sender retransmits if an error occurs
- ④ Error Handling : Detects errors (e.g. lost / missing data) and ensures complete and correct file transfer
- ⑤ User services : Hides underlying network details and provides a simple, reliable interface to users
- ⑥ Supports recovery : If a transfer is interrupted, FTP can resume from the last successful point

Q8) Given,

Application layer delay = 4 ms

Transport layer delay = 3 ms

Network layer delay = 5 ms

Data link layer delay = 2 ms

Physical layer delay = 1 ms

Transmission delay = 12 ms

Propagation delay = 18 ms

$$\begin{aligned}
 \text{Total delay} &= \text{Delay at each layer added together} \\
 &= 4 \text{ ms} + 3 \text{ ms} + 5 \text{ ms} + 2 \text{ ms} + 1 \text{ ms} \\
 &= 15 \text{ ms}
 \end{aligned}$$

PROCESSING DELAYS AT OSI LAYERS

- ① APPLICATION LAYER : The data is processed for user applications such as file transfer, email, web services, etc...
- ② TRANSPORT LAYER : Performs tasks like segmentation, error checking, flow control
- ③ NETWORK LAYER : Handles logical addressing, routing, and packet forwarding
- ④ DATA LINK LAYER : Responsible for framing, MAC addressing, and error detection
- ⑤ PHYSICAL LAYER : Converts data into bits and transmits over the physical medium

Q9) Given,

$$\text{no. of. routers} = 3 \quad (R_1, R_2, R_3)$$

$$\text{Processing delays} = 2 \text{ ms}, 3 \text{ ms}, 2 \text{ ms}$$

$$\text{Propagation delays} = 15 \text{ ms}, 20 \text{ ms}, 10 \text{ ms}$$

$$\text{Packet size} = 2 \text{ KB} \rightarrow 2 \times 1024 \text{ Bytes} = 2048 \text{ Bytes}$$

$$\text{Link Bandwidth} = 2 \text{ Mbps}$$

TOTAL PROCESSING DELAY = $2 + 3 + 2 \rightarrow 7 \text{ ms}$

TRANSMISSION DELAY = $\frac{16384}{2 \times 10^6} \rightarrow 0.008192$
= $8.192 \text{ ms (per link)}$

PROPAGATION DELAY = $15 + 20 + 10 \rightarrow 45 \text{ ms}$

TOTAL DELAY = processing delay + Transmission delay + Propagation
= $7 \text{ ms} + 24.576 \text{ ms} + 45 \text{ ms}$
= 76.576 ms

ROLE OF INTERMEDIATE ROUTERS IN PACKET FORWARDING :

PATH DETERMINATION : Routers examine the destination IP address and consult their routing table to determine the next hop

TRAFFIC MANAGEMENT : Routers manage traffic by queuing, scheduling and avoiding congestion

ERROR HANDLING : Routers can detect problems (eg: link failure) and send ICMP messages back to the source

Q10) Given ,

No. of routers = 2 (R_1 & R_2)

Processing delay at R_1 = 3 ms

Processing delay at R_2 = 4 ms

TOTAL PROCESSING DELAY = $3\text{ ms} + 4\text{ ms} \rightarrow 7\text{ ms}$

TRANSMISSION DELAY =
$$\frac{\text{Packet size (bits)}}{\text{Bandwidth}}$$

Packet size = $1500\text{ Bytes} \rightarrow 12000\text{ bits}$

Transmission delay =
$$\frac{12000}{1.5 \times 10^6} = 0.008\text{ s} \rightarrow 8\text{ ms}$$

PROPAGATION DELAY = $12 + 15 + 10 \rightarrow 37\text{ ms}$

TOTAL DELAY = Processing delay + Transmission delay + propagation
= $7\text{ ms} + 24\text{ ms} + 37\text{ ms}$
= 68 ms

PACKET FORWARDING PROCESS ACROSS LAYERS

APPLICATION :

- Creates data for transmission
- Delivers data to application (at destination)

TRANSPORT :

- Segments data, adds ports, ensures reliability
- Router not processed (at destination)

NETWORK :

- Adds IP header (source IP, Dest IP)
- Accepts packet if destination IP matches

 DATA LINK :

- Adds MAC header and trailer, creates frame
- Removes frame header, checks FCS (at destination)

Q ₁	Q ₂	Q ₃	Q ₄	Q ₅	Q ₆	Q ₇	Q ₈	Q ₉	Q ₁₀
131211	131211	131211	131211	131211	131211	131211	131211	131211	131211

Understandings 2.25 2.25 2.25 2.25 2.25 2.25 2.25 2.25 2.25 2.25

45%

Explanation 35%

clarity 20%

4.25 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5 4.5

$$\frac{41.75}{50} = 83.5\%$$